> **IMPLEMENTATION NOTE**
>
> CXL components shall interpret the PCIe MMB Command Opcode Vendor ID = 1E98h or 0000h with CXL defined commands. 0000h is a PCI-SIG reserved value for legacy CXL compatibility. However, it is strongly recommended for callers to use the CXL Vendor ID (1E98h) to identify CXL defined commands.

8.2.10.1 Information and Status Command Set

The Information and Status command set includes commands for querying component capabilities and status.

8.2.10.1.1 Identify (Opcode 0001h)

Retrieves status information about the component, including whether it is ready to process commands. A component that is not ready to process commands shall return 'Retry Required'.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required

Command Effects:

- None

Table 8-50. Identify Output Payload (Sheet 1 of 2)

Byte Offset	Length in Bytes	Description
00h	2	**PCIe Vendor ID**: Identifies the manufacturer of the component, as defined in PCIe Base Specification.
02h	2	**PCIe Device ID**: Identifier for this particular component assigned by the vendor, as defined in PCIe Base Specification.
04h	2	**PCIe Subsystem Vendor ID**: Identifies the manufacturer of the subsystem, as defined in PCIe Base Specification.
06h	2	**PCIe Subsystem ID**: Identifier for this particular subsystem assigned by the vendor, as defined in PCIe Base Specification.

Table 8-50. Identify Output Payload (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
08h	8	**Device Serial Number:** Unique identifier for this device, as defined in the Device Serial Number Extended Capability in PCIe Base Specification.
10h	1	**Maximum Supported Message Size:** The maximum supported size of the Message Payload (as defined in Figure 7-19) in bytes for any requests sent to this component, expressed as 2^n. The minimum supported size is 256 bytes (n=8) and the maximum supported size is 1024 bytes (n=10). This field is used by the caller to limit the Message Payload size such that the size of the Message Body does not exceed the capabilities of the component. The component shall discard any received messages that exceed the maximum Message Payload size advertised in this field in a manner that prevents any internal receiver hardware errors. The component shall return a response message with the 'Invalid Payload Length' return code for all received request messages with Message Payload Length that exceeds the maximum size advertised in this field. The CXL specification guarantees that the size of the Identify Output Payload shall never exceed 256 bytes.
11h	1	**Component Type:** Indicates the type of component. • 00h = Switch • 03h = Type 3 Device • 04h = GFD • All other encodings are reserved

8.2.10.1.2 Background Operation Status (Opcode 0002h)

Retrieve information about outstanding Background Operations processing on the interface from which this command was received.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

Table 8-51. Background Operation Status Output Payload

Byte Offset	Length in Bytes	Description
00h	1	**Background Operation Status:** Reports the status of outstanding Background Operations. • Bit[0]: **Background Operation**: Indicates whether a background operation is in progress, as defined in Section 8.2.9.4.6. • Bits[7:1]: **Percentage Complete**: The percentage complete (0-100) of the background command, as defined in Section 8.2.9.4.7.
01h	1	**Reserved**
02h	2	**Command Opcode:** The command identifier of the last command executed in the background. See Section 8.2.10 for the list of command opcodes.
04h	2	**Return Code:** The result of the command run in the background. Only valid when Percentage Complete = 100. See Section 8.2.9.4.5.1.
06h	2	**Vendor Specific Extended Status:** The vendor specific extended status of the last background command. Valid only when Percentage Complete = 100.

8.2.10.1.3 Get Response Message Limit (Opcode 0003h)

Retrieves the current configured response message limit used by the component. The component shall limit the output payload size of commands with variably sized outputs such that the Message Payload does not exceed the value reported by this command.

Possible Command Return Codes:

- Success
- Internal Error
- Retry Required

Command Effects:

- None

Table 8-52. Get Response Message Limit Output Payload

Byte Offset	Length in Bytes	Description
00h	1	**Response Message Limit**: The configured maximum size of the Message Payload for the response message generated by the component (as defined in Figure 7-19) in bytes, expressed as 2^n.307.32 pt The minimum supported size is 256 bytes (n=8) and the maximum supported size is 1024 bytes (n=10).

8.2.10.1.4 Set Response Message Limit (Opcode 0004h)

Configures the response message limit used by the component. The component shall limit the output payload size of commands with variably sized outputs such that the Message Payload does not exceed the value programmed with this command. Components shall return "Internal Error" if any errors within the component prevent it from limiting its response message size to a value lower than or equal to the requested size.

Possible Command Return Codes:

- Success
- Invalid Input
- Internal Error
- Retry Required
- Invalid Payload Length

Command Effects:

- None

Table 8-53. Set Response Message Limit Input Payload

Byte Offset	Length in Bytes	Description
00h	1	**Response Message Limit**: The configured maximum size of the Message Payload for response messages generated by the component (as defined in Figure 7-19) in bytes, expressed as 2^n. The minimum supported size is 256 bytes (n=8) and the maximum supported size is 1024 bytes (n=10).

Table 8-54. Set Response Message Limit Output Payload

Byte Offset	Length in Bytes	Description
00h	1	**Response Message Limit**: The configured maximum size of the Message Payload for response messages generated by the component (as defined in Figure 7-19) in bytes, expressed as 2^n. The value returned is the Response Message Limit used by the component after processing this request, which may be less than the requested value. The minimum supported size is 256 bytes (n=8) and the maximum supported size is 1024 bytes (n=10). The FM shall discard any messages sent by the component with Message Payload Length that exceed the maximum size set with this field in a manner that prevents any internal receiver hardware errors.

8.2.10.1.5 Request Abort Background Operation (Opcode 0005h)

Request Abort Background Operation requests the device to abort the ongoing background operation on the same CCI. In response, the device may choose to interrupt the ongoing background operation or may choose to continue its execution until completion. If the background operation is interrupted, a Command Return code of Success shall be returned, the Percentage Complete field in the Background Command Status register shall indicate a value smaller than 100, and the Background Operation field in the Mailbox Status register shall be cleared. If the ongoing background operation does not support abort capability, the Request Abort Background Operation return code shall be set to Request Abort Not Supported by Background Operation. For this command, both the Input Payload and Output Payload are empty.

Possible Command Return Codes:

- Success
- Unsupported
- Internal Error
- Request Abort Not Supported by Background Operation

Command Effects:

- None

8.2.10.2 Events

This section defines the standard event record format that all CXL devices shall use when reporting events to the host. Also defined are the Get Event Record and Clear Event Record commands that operate on those event records. The device shall support at least 1 event record within each event log. Devices shall return event records to the host in the temporal order the device detected the events in. The event occurring the earliest in time, in the specific event log, shall be returned first.

8.2.10.2.1 Event Records

This section describes the events reported by devices through the Get Event Records command. The device shall use the Common Event Record format when generating events for any event log.

A CXL memory device that implements the PCIe Configuration Space Header Class Code defined in Section 8.1.12.1 or advertises Memory Device Command support in the Mailbox Capabilities register (see Section 8.2.9.4.3) shall use the Memory Module Event Record format when reporting general device events and shall use either the General Media Event Record or DRAM Event Record when reporting media events.

Table 8-55. Common Event Record Format (Sheet 1 of 2)

Byte Offset	Length in Bytes	Description
00h	10h	**Event Record Identifier**: UUID representing the specific Event Record format. The following UUIDs are defined in this spec: • fbcd0a77-c260-417f-85a9-088b1621eba6 – General Media Event Record (see Table 8-57) • 601dcbb3-9c06-4eab-b8af-4e9bfb5c9624 – DRAM Event Record (see Table 8-58) • fe927475-dd59-4339-a586-79bab113b774 – Memory Module Event Record (see Table 8-59) • e71f3a40-2d29-4092-8a39-4d1c966c7c65 - Memory Sparing Event Record (see Table 8-60) • 77cf9271-9c02-470b-9fe4-bc7b75f2da97 – Physical Switch Event Record (see Table 7-77) • 40d26425-3396-4c4d-a5da-3d47263af425 – Virtual Switch Event Record (see Table 7-78) • 8dc44363-0c96-4710-b7bf-04bb99534c3f – MLD Port Event Record (see Table 7-79) • ca95afa7-f183-4018-8c2f-95268e101a2a - Dynamic Capacity Event Record (see Table 8-62)
10h	1	**Event Record Length**: Number of valid bytes that are in the event record, including all fields.
11h	3	**Event Record Flags**: Multiple bits may be set. • Bits[1:0]: **Event Record Severity**: The severity of the event. — 00b = Informational Event — 01b = Warning Event — 10b = Failure Event — 11b = Fatal Event • Bit[2]: **Permanent Condition**: The event reported represents a permanent condition for the device. This shall not be set when reporting Event Record Severity of Informational. • Bit[3]: **Maintenance Needed**: The device requires maintenance. This shall not be set when reporting Event Record Severity of Informational. • Bit[4]: **Performance Degraded**: The device is no longer operating at optimal performance. This shall not be set when reporting Event Record Severity of Informational. • Bit[5]: **Hardware Replacement Needed**: The device should immediately be replaced. This shall not be set when reporting Event Record Severity of Informational. If this bit is set and the Component Identifier field in the event record is valid, the hardware to be replaced is the hardware identified by the Component Identifier field. If this bit is set and the Component Identifier field in the event record is invalid or not part of the event record, the hardware to be replaced is the entire device. • Bit[6]: **Maintenance Operation Subclass Valid Flag**: If set, the Maintenance Operation Subclass is valid. This bit applies only to CXL devices. • Bits[7]: **LD-ID Valid Flag**. If set to 1, the LD-ID field is valid and this event is applicable to only the specified LD. If set to 0, and the Head ID Valid Flag is also 0, the event is applicable to the entire device. If set to 0, and the Head ID Valid Flag is 1, the event is applicable the entire device head. • Bits[8]: **Head ID Valid Flag**. If set to 1, the Head ID field is valid. MH-SLDs and MH-MLDs shall set this value to 1 if setting the LD-ID Valid Flag to 1. • Bits[23:9]: **Reserved**
14h	2	**Event Record Handle**: The event log unique handle for this event record. This is the value that the host shall use when requesting the device to clear events using the Clear Event Records command. This value shall be nonzero.
16h	2	**Related Event Record Handle**: Optional event record handle to another related event in the same event log. If there are no related events, this field shall be cleared to 0000h.
18h	8	**Event Record Timestamp**: The time the device recorded the event. The number of unsigned nanoseconds that have elapsed since midnight, 01-Jan-1970, UTC. If the device does not have a valid timestamp, return all 0s.

Table 8-55. Common Event Record Format (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
20h	1	**Maintenance Operation Class**: This field indicates the maintenance operation the device requests to initiate. If the device does not have any requests, this field shall be cleared to 00h. This field applies only to CXL devices. See Table 8-125.
21h	1	**Maintenance Operation Subclass**: This field indicates the maintenance operation subclass that the device recommends to initiate. If the device does not have a specific recommendation for the subclass, the Maintenance Operation Subclass Valid flag shall be 0. This field applies only to CXL devices. See Table 8-125.
22h	2	**LD-ID**: LD ID of LD from where the event originated. Only valid if LD-ID valid flag is set to 1.
24h	1	**Head ID**: ID of the Device head, from where the event originated. Only valid if head valid flag is set to 1.
25h	0Bh	**Reserved**
30h	50h	**Event Record Data**: Format depends on the Event Record Identifier

When reporting events using per-LD level Primary Mailbox or Secondary Mailbox CCI instances, the event consumer considers that LD as an independent device regardless of whether the LD is an SH-SLD, or one of multiple LDs in an SH-MLD, or LD is from an MH-SLD, or LD is from one of multiple LDs in an MH-MLD. As such, it is recommended that the device clear LD-ID Valid Flag to 0 and Head ID Valid Flag to 0, thereby not reporting any LD-ID or Head ID details.

When reporting events using CCI instances which are potentially common to multiple LDs (e.g. MCTP based CCI instances), the event consumer will be aware of all LDs discoverable through that CCI instance. As such, it is recommended that the device follows the below guidelines for usage of the LD-ID Valid Flag, Head ID Valid Flag, LD-ID, and Head ID fields to inform the event consumer about the origin and scope of the reported events.

For SH-SLDs:
The device should choose to clear the LD-ID Valid Flag and Head ID Valid Flag to '0' indicating the LD-ID and Head ID fields are invalid. The event consumer would have already discovered that the device is an SH-SLD as part of the device discovery, initialization, and configuration process; hence, the event consumer already has head and LD association.

For SH-MLDs:
The device should choose to keep the Head ID Valid Flag set to '0' indicating the Head ID field is invalid, as the event consumer would have already discovered that the device is a Single Headed Device as part of the device discovery, initialization, and configuration process; hence, the event consumer already has head association.

If the given event impacts only one LD out of all LDs then the LD-ID Valid Flag should be set to '1' and LD-ID field should contain the LD ID of the impacted LD. If the given event impacts more than one LDs, the device should send multiple events (one for every impacted LD) with the LD-ID Valid Flag set to '1' and the LD-ID field indicating the impacted LD; additionally using "Related Event Record Handle" and "Event Record Timestamp" fields to link all related events together. Separate event records should be generated if the event applies to a specific physical memory location that is shared across multiple LDs, and that physical memory location is represented using different DPAs on those LDs. If the device is impacted by an event which impacts globally to all LDs, the device should choose to send the event with the LD-ID Valid Flag set to '0' indicating the LD-ID field is invalid and the event impacts the entire device.

For MH-SLDs and MH-MLDs:
If the given event impacts only one head out of all heads the Head-ID Valid Flag should be set to '1' and the Head ID field should contain the Head ID of the impacted head. If the given event impacts more than one heads, the device should send multiple events (one for every impacted head) with the Head ID Valid Flag set to '1' and the Head ID field indicating the impacted head; additionally using "Related Event Record Handle" and "Event Record Timestamp" fields to link all related events together. Separate event records should be generated if the event applies to a specific physical memory location that is shared across multiple LDs across multiple Heads, and that physical memory location is represented using different DPAs on those LDs across multiple Heads. If the device is impacted by an event which impacts globally to all heads, the device should choose to send the event with the Head ID Valid Flag set to '0' indicating the Head ID field is invalid and the event impacts the entire device.

Additionally, if the given event impacts only one LD out of all LDs on a given head, the LD-ID Valid Flag should be set to '1' and the LD-ID field should contain the LD-ID of the impacted LD. If the given event impacts more than one LDs on a given head, the device should send multiple events (one for every impacted LD) with the LD-ID Valid Flag set to '1' and the LD-ID field indicating the impacted LD; additionally using "Related Event Record Handle" and "Event Record Timestamp" fields to link all related events together. Separate event records should be generated if the event applies to a specific physical memory location that is shared across multiple LDs on given head, and that physical memory location is represented using different DPAs on those LDs on given head. If the device is impacted by an event which impacts globally to all LDs on a given head, the device should choose to send the event with LD-ID Valid Flag set to '0' indicating the LD-ID field is invalid and the event impacts all LDs on the given head.

Please note that in case of MH-SLDs and MH-MLDs, if the device wants to report an event while setting the LD-ID Valid Flag to 1, the device must set the Head ID Valid Flag to '1' and provide a valid Head ID.

A number of Event Record formats include a Component Identifier Field. The format of this field is defined in Table 8-56.

Table 8-56. Component Identifier Format

Byte Offset	Length in Bytes	Description
00h	1	**ID Validity Flags**: Indicators of which fields are valid. • Bit[0]: If set, PLDM Entity Identification Information is valid • Bit[1]: If set, PLDM Resource ID is valid • Bits[7:2]: **Reserved**
01h	15h	If ID Validity Flags[1:0]=00b: • Bytes[14:0] are reserved If ID Validity Flags[1:0]=01b: • Bytes[5:0] = PLDM Entity Identification Information for the lowest-level entity impacted • Bytes[14:6] are reserved If ID Validity Flags[1:0]=10b: • Bytes[5:0] are reserved • Bytes[9:6] = Resource ID for the lowest-level entity impacted • Bytes[14:10] are reserved. If ID Validity Flags[1:0]=11b: • Bytes[5:0] = PLDM Entity Identification Information for the lowest-level entity impacted • Bytes[9:6] = Resource ID for the lowest-level entity impacted • Bytes[14:10] are reserved The format of the Entity Identification Information (including the endianness) is defined in the Platform-Level Data Model (PLDM) for Platform Monitoring and Control Specification. The format of (including the endianness) is also defined in the PLDM for Platform Monitoring and Control Specification.

IMPLEMENTATION NOTE

The Component ID field points to the lowest isolated entity of a device's hierarchical model. When the Component ID format is governed by this specification, Entity Identification Information and/or Resource ID are handled as follows:

- If a CXL Type 3 Device has field-replaceable sub-components such as DIMMs and the error is associated with a specific DRAM device, the Entity Identification Information and/or Resource ID for that DRAM device is provided.
- If a specific DIMM on a CXL Type 3 Device fails training, but the device does not know which DRAM device on that DIMM is the source of the error, the Entity Identification Information and/or Resource ID for the failing DIMM is provided.
- If a CXL Type 3 Device reports an event, but it is unable to isolate to a specific sub-component or sub-FRU, then the Entity Identification Information and/or Resource ID for that CXL device is provided.

8.2.10.2.1.1 General Media Event Record

The General Media Event Record defines a general media related event. The device shall generate a General Media Event Record for each general media event occurrence. Unless specifically stated otherwise, event record content shall be generated with information on what triggered the event. For example, when a corrected error threshold is exceeded, the memory address and transaction type for the actual corrected error at the time of being exceeded shall be used.

If a General Media event occurs on multiple identifiable components related to the same cacheline, and if the Component Identifier field is valid, a General Media Event Record shall be generated for each component. Additionally, the Related Event Record Handle field of the Common Event Record (see Table 8-55) shall be used to describe such relationships.

The layout of a General Media event record is shown in Table 8-57.

Table 8-57. General Media Event Record (Sheet 1 of 3)

Byte Offset	Length in Bytes	Description
00h	30h	**Common Event Record**: See corresponding common event record fields defined in Section 8.2.10.2.1. The Event Record Identifier field shall be set to fbcd0a77-c260-417f-85a9-088b1621eba6 which identifies a General Media Event Record.
30h	8	**Physical Address**: The physical address where the memory event occurred. • Bit[0]: **Volatile**: When set, indicates the DPA field is within the volatile memory range. When cleared, indicates the DPA is within the persistent memory range. • Bit[1]: **DPA not repairable**.[1] • Bits[5:2]: **Reserved**. • Bits[7:6]: **DPA[7:6]**. • Bits[15:8]: **DPA[15:8]**. • ... • Bits[63:56]: **DPA[63:56]**.
38h	1	**Memory Event Descriptor**: Additional memory event information. Unless specified below, these shall be valid for every Memory Event Type reported. • Bit[0]: **Uncorrectable Event**: When set, indicates the reported event is uncorrectable by the device. When cleared, indicates the reported event was corrected by the device. • Bit[1]: **Threshold Event**: When set, the event is the result of a threshold on the device having been reached. When cleared, the event is not the result of a threshold limit. • Bit[2]: **Poison List Overflow Event**: When set, the Poison List has overflowed, and this event is not in the Poison List. When cleared, the Poison List has not overflowed. • Bits[7:3]: **Reserved**
39h	1	**Memory Event Type**: Identifies the type of event that occurred. The specific memory event types logged by the device will depend on the RAS mechanisms implemented in the device and is implementation dependent. • 00h = Media ECC Error. ECC error discovered during normal operations or background operations other than scrub operations. • 01h = Invalid Address. A host access was for an invalid address range. The DPA field shall contain the invalid DPA the host attempted to access. When returning this event type, the Poison List Overflow Event descriptor does not apply. • 02h = Data Path Error. Internal device data path, media link, or internal device structure errors not directly related to the media. • 03h = TE State Violation. The device detected a TE State access control violation during a memory request. Access control checks, device behavior, and logging are described in Section 11.5. • 04h = Scrub Media ECC Error. ECC error discovered during scrub operations. • 05h = Advanced Programmable Corrected Memory Error Counter Expiration. Indicates that corrected memory error counters used by an Advanced Programmable Threshold Feature have expired and reset. If this event type is set, the Threshold Event bit in the Memory Event Descriptor shall be set. • 06h = CKID Violation. The device detected a CKID access control violation during a memory request. Access control checks, device behavior, and logging are described in Section 11.5. • All other encodings are reserved. **Note:** The ECC errors event types listed above are detected by the memory media or memory controller.

Table 8-57. General Media Event Record (Sheet 2 of 3)

Byte Offset	Length in Bytes	Description
3Ah	1	**Transaction Type**: The first device detected transaction that caused the event to occur. • 00h = Unknown/Unreported. • 01h = Host Read. • 02h = Host Write. • 03h = Host Scan Media. • 04h = Host Inject Poison. • 05h = Media Patrol Scrub. • 06h = Media Management. • 07h = Internal Media Error Check Scrub. • 08h = Media Initialization. Media training and testing before Memory_Active or Memory_Active_Degraded is set. • All other encodings are reserved.
3Bh	2	**Validity Flags**: Indicators of which fields are valid in the returned data. • Bit[0]: When set, the Channel field is valid • Bit[1]: When set, the Rank field is valid • Bit[2]: When set, the Device field is valid • Bit[3]: When set, the Component Identifier field is valid • Bit[4]: When bit[3] and bit[4] are both set, the format of the Component Identifier field is governed by Table 8-56 • Bits[15:5]: **Reserved**
3Dh	1	**Channel**: The channel of the memory event location. A channel is defined as an interface that can be independently accessed for a transaction. The CXL device may support one or more channels.
3Eh	1	**Rank**: The rank of the memory event location. A rank is defined as a set of memory devices on a channel that together execute a transaction. Multiple ranks may share a channel.
3Fh	3	**Device**: Bitmask that represents all devices in the rank associated with the memory event location.
42h	10h	**Component Identifier**: Device-specific component identifier for the event. This may describe a field-replaceable sub-component of the device. This field is valid when bit[3] of the Validity Flags field is set. When bit[4] of the Validity Flags field is set, the component identifier is defined by Table 8-56; otherwise, the component identifier format is device specific.
52h	1	**Advanced Programmable Corrected Memory Error Threshold Event Flags**: Field is valid if the Threshold Event bit is set in the Memory Event Descriptor field and an Advanced Programmable Threshold Feature is enabled. • Bit[0]: The device has detected corrected memory errors in more than one memory media component before the counter has expired • Bit[1]: The programmable threshold has been exceeded • Bits[7:2]: **Reserved**

Table 8-57. General Media Event Record (Sheet 3 of 3)

Byte Offset	Length in Bytes	Description
53h	3	**Corrected Memory Error Count at Event**: Accumulated corrected memory error count that is used by an Advanced Programmable Corrected Memory Error Threshold Feature at the time of the event. This field is valid if the Threshold Event bit is set in the Memory Event Descriptor field and an Advanced Programmable Corrected Memory Error Threshold Feature is enabled. For events in which the advanced programmable threshold is exceeded, this field value shall be equal to the threshold that is exceeded. For events in which the advanced programmable threshold counter has expired, this field value shall be a value greater than 0. Counter expiration events in which the corrected memory error count is 0 shall not generate a media event record.
56h	1	**Memory Event Sub-Type**: Identifies the sub-type of the Memory Event Type that occurred. If Memory Event Type = 02h (Data Path Error), this field is defined as follows: • 00h = Memory Event Sub-Type Not Reported • 01h = Internal Datapath Error • 02h = Media Link Command Training Error • 03h = Media Link Control Training Error • 04h = Media Link Data Training Error • 05h = Media Link CRC Error • All other encodings are reserved Other Memory Event Type values: • 00h = Memory Event Sub-Type Not Reported • All other encodings are reserved
57h	29h	**Reserved**

1. If the DPA needs to be repaired but there is no resource available, then bit[1] of Physical Address field shall be set to 1 and Maintenance Needed bit in Event Record Flags shall be cleared to 0.

8.2.10.2.1.2 DRAM Event Record

The DRAM Event Record defines a DRAM-related event. The device shall generate a DRAM Event Record for each DRAM event occurrence. The layout of a DRAM Event Record is shown Table 8-58. Unless specifically stated otherwise, event record content shall be generated with information on what triggered the event. For example, when a corrected error threshold is exceeded, the memory address and transaction type for the actual corrected error at the time of being exceeded shall be used.

If a DRAM event occurs on multiple components related to the same cacheline, and if the Component Identifier field is valid, a DRAM Event Record shall be generated for each component. Additionally, the Related Event Record Handle field of the Common Event Record (see Table 8-55) shall be used to describe such relationships.

Table 8-58. DRAM Event Record (Sheet 1 of 3)

Byte Offset	Length in Bytes	Description
00h	30h	**Common Event Record**: See corresponding common event record fields defined in Section 8.2.10.2.1. The Event Record Identifier field shall be set to 601dcbb3-9c06-4eab-b8af-4e9bfb5c9624 which identifies a DRAM Event Record.
30h	8	**Physical Address**: The physical address where the memory event occurred. • Bit[0]: **Volatile**: When set, indicates the DPA field is within the volatile memory range. When cleared, indicates the DPA is within the persistent memory range. • Bit[1]: **DPA not repairable**.[1] • Bits[5:2]: **Reserved**. • Bits[7:6]: **DPA[7:6]**. • Bits[15:8]: **DPA[15:8]**. • ... • Bits[63:56]: **DPA[63:56]**.
38h	1	**Memory Event Descriptor**: Additional memory event information. Unless specified below, these shall be valid for every Memory Event Type reported. • Bit[0]: **Uncorrectable Event**: When set, indicates the reported event is uncorrectable by the device. When cleared, indicates the reported event was corrected by the device. • Bit[1]: **Threshold Event**: When set, the event is the result of a threshold on the device having been reached. When cleared, the event is not the result of a threshold limit. • Bit[2]: **Poison List Overflow Event**: When set, the Poison List has overflowed, and this event is not in the Poison List. • Bits[7:3]: **Reserved**.
39h	1	**Memory Event Type**: Identifies the type of event that occurred. The specific memory event types logged by the device will depend on the RAS mechanisms implemented in the device and is implementation dependent. • 00h = Media ECC Error. ECC error discovered during normal operations or background operations other than scrub operations. • 01h = Scrub Media ECC Error. ECC error discovered during scrub operations. • 02h = Invalid Address. A host access was for an invalid address. The DPA field shall contain the invalid DPA the host attempted to access. When returning this event type, the Poison List Overflow Event descriptor does not apply. • 03h = Data Path Error. Internal device data path, media link, or internal device structure errors not directly related to the media. • 04h = TE State Violation. The device detected a Non-TEE access to TEE Exclusive memory or a TEE access to Non-TEE Exclusive memory and has returned all 1s if the Transaction Type is a Host Read or dropped the transaction if the Transaction Type is a Host Write of a partial cacheline. Memory events of this type shall utilize the Informational Event Log and report an Event Record Severity of Informational Event. This is described in Section 11.5. • 05h = Advanced Programmable Corrected Memory Error Counter Expiration. Indicates that corrected memory error counters used by an Advanced Programmable Threshold Feature have expired and reset. If this event type is set, the Threshold Event bit in the Memory Event Descriptor shall be set. • 06h = CKID Violation. The device detected a CKID value in the memory transaction that was outside the device's configured range, or the memory transaction CKID referenced an OSCKID utilizing a TEE opcode, or the memory transaction CKID referenced a TVMCKID utilizing a non-TEE opcode. Events of this type shall utilize the Informational Event Log and report an Event Record Severity of Informational Event. This is described in Section 11.5. • All other encodings are reserved. **Note:** The ECC errors event types listed above are detected by the memory media or memory controller.

Table 8-58. DRAM Event Record (Sheet 2 of 3)

Byte Offset	Length in Bytes	Description
3Ah	1	**Transaction Type:** The first device detected transaction that caused the event to occur. • 00h = Unknown/Unreported. • 01h = Host Read. • 02h = Host Write. • 03h = Host Scan Media. • 04h = Host Inject Poison. • 05h = Media Patrol Scrub. • 06h = Media Management. • 07h = Internal Media Error Check Scrub. • 08h = Media Initialization. Media training and testing before Memory_Active or Memory_Active_Degraded is set. • All other encodings are reserved.
3Bh	2	**Validity Flags:** Indicators of which fields are valid in the returned data. • Bit[0]: When set, the Channel field is valid • Bit[1]: When set, the Rank field is valid • Bit[2]: When set, the Nibble Mask field is valid • Bit[3]: When set, the Bank Group field is valid • Bit[4]: When set, the Bank field is valid • Bit[5]: When set, the Row field is valid • Bit[6]: When set, the Column field is valid • Bit[7]: When set, the Correction Mask field is valid • Bit[8]: When set, the Component Identifier field is valid • Bit[9]: When bit[8] and bit[9] are both set, the format of the Component Identifier field is governed by Table 8-56 • Bit[10]: When set, the Sub-channel field is valid • Bits[15:11]: **Reserved**
3Dh	1	**Channel:** The channel of the memory event location. A channel is defined as an interface that can be independently accessed for a transaction. The CXL device may support one or more channels.
3Eh	1	**Rank:** The rank of the memory event location. A rank is defined as a set of memory devices on a channel that together execute a transaction. Multiple ranks may share a channel.
3Fh	3	**Nibble Mask:** Identifies one or more nibbles in error on the memory bus producing the event. Nibble Mask bit 0 shall be set if nibble 0 on the memory bus produced the event, etc. This field should be valid for corrected memory errors. See the implementation note following this table for an example of how this field is intended to be used.
42h	1	**Bank Group:** The bank group of the memory event location
43h	1	**Bank:** The bank number of the memory event location
44h	3	**Row:** The row number of the memory event location
47h	2	**Column:** The column number of the memory event location
49h	20h	**Correction Mask:** Identifies the bits in error within that nibble in error on the memory bus that is producing the event. The lowest nibble in error in the Nibble Mask uses Correction Mask 0, the next-lowest nibble uses Correction Mask 1, etc. Burst position 0 uses Correction Mask nibble 0, etc. Four correction masks allow for up to 4 nibbles in error. This field should be valid for corrected memory errors. See the implementation note following this table for an example of how this field is intended to be used. • Offset 49h: **Correction Mask 0** (8 bytes) • Offset 51h: **Correction Mask 1** (8 bytes) • Offset 59h: **Correction Mask 2** (8 bytes) • Offset 61h: **Correction Mask 3** (8 bytes)

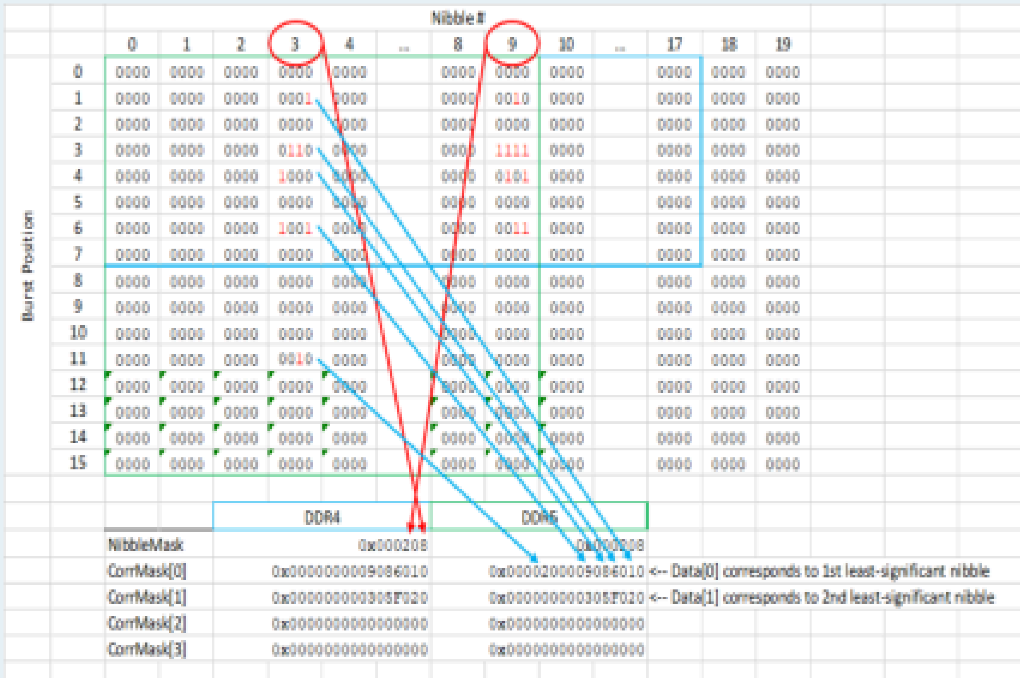
Table 8-58. DRAM Event Record (Sheet 3 of 3)

Byte Offset	Length in Bytes	Description
69h	10h	**Component Identifier**: Device-specific component identifier for the event. This may describe a field-replaceable sub-component of the device. This field is valid when bit[8] of the Validity Flags field is set. When bit[9] of the Validity Flags field is set, the component identifier is defined by Table 8-56; otherwise, the component identifier format is device specific.
79h	1	**Sub-channel:** The sub-channel of the memory event location.
7Ah	1	**Advanced Programmable Corrected Memory Error Threshold Event Flags**: This field is valid if the Threshold Event bit is set in the Memory Event Descriptor Field and the Advanced Programmable CVME Threshold Feature is enabled. • Bit[0]: The device has detected corrected memory errors in more than one memory media component before the counter has expired • Bit[1]: The programmable threshold has been exceeded • Bits[7:2]: **Reserved**
7Bh	3	**CVME Count at Event:** Accumulated CVME count used by the Advanced Programmable CVME Threshold Feature at the time of the event. This field is valid when the Threshold Event bit is set in the Memory Event Descriptor field and the Advanced Programmable CVME Threshold Feature is enabled. For events in which the advanced programmable threshold is exceeded, this field value shall be equal to the threshold that is exceeded. For events in which the advanced programmable threshold counter has expired, this field value shall be a value greater than 0. Counter expiration events in which the CVME count is 0 shall not generate a DRAM event record.
7Eh	1	**Memory Event Sub-Type**: Identifies the sub-type of Memory Event Type that occurred. If Memory Event Type = 03h (Data Path Error), this field is defined as follows: • 00h = Memory Event Sub-Type Not Reported • 01h = Internal Datapath Error • 02h = Media Link Command Training Error • 03h = Media Link Control Training Error • 04h = Media Link Data Training Error • 05h = Media Link CRC Error • All other encodings are reserved Other Memory Event Type values: • 00h = Memory Event Sub-Type Not Reported • All other encodings are reserved
7Fh	1	**Reserved**

1. If the DPA needs to be repaired but there is no resource available, then bit[1] of the Physical Address field shall be set to 1 and the Maintenance Needed bit in Event Record Flags shall be cleared to 0.

IMPLEMENTATION NOTE

The following example illustrates how the Nibble Mask and Correction Mask are used for a sample DDR4 and DDR5 DRAM implementation behind a CXL memory device where nibble #3 and #9 contain the location of the corrected error.



8.2.10.2.1.3 Memory Module Event Record

The layout of a Memory Module Event Record is shown in Table 8-59.

Table 8-59. Memory Module Event Record

Byte Offset	Length in Bytes	Description
00h	30h	**Common Event Record:** See corresponding common event record fields defined in Section 8.2.10.2.1. The Event Record Identifier field shall be set to fe927475-dd59-4339-a586-79bab113b774 which identifies a Memory Module Event Record.
30h	1	**Device Event Type:** Identifies the type of event that occurred. The specific device event types logged by the device will depend on the RAS mechanisms implemented in the device and is implementation dependent. • 00h = Health Status Change. • 01h = Media Status Change. • 02h = Life Used Change. • 03h = Temperature Change. • 04h = Data Path Error. Internal device data path, media link or internal device structure errors not directly related to the media, training, or CRC. • 05h = LSA Error. An error occurred in the device Label Storage Area. • 06h = Unrecoverable Internal Sideband Bus Error. • 07h = Memory Media FRU Error. • 08h = Power Management Fault. • All other encodings are reserved.
31h	12h	**Device Health Information:** A complete copy of the device's health info at the time of the event. The format of this field is described in Table 8-148.
43h	2	**Validity Flags:** Indicators of which fields within the record are valid. • Bit[0]: When set, the Component Identifier field is valid • Bit[1]: When bit[0] and bit[1] are both set, the format of the Component Identifier field is governed by Table 8-56 • Bits[15:2]: **Reserved**
45h	10h	**Component Identifier:** Device-specific component identifier for the event. This may describe a field-replaceable sub-component of the device. This field is valid when bit[0] of the Validity Flags field is set. When bit[1] of the Validity Flags field is set, the component identifier is defined by Table 8-56; otherwise, the component identifier format is device specific.
55h	1	**Device Event Sub-Type:** Identifies the sub-type of Device Event Type that occurred. If Device Event Type = 07h (Memory Media FRU Error), this field is defined as follows: • 00h = Device Event Sub-Type Not Reported. • 01h = Invalid Configuration Data. Configuration data (e.g., SPD) for memory media FRU is invalid or corrupt. • 02h = Unsupported Configuration Data. Configuration data (e.g., SPD) for memory media FRU is unsupported or unrecognized. • 03h = Unsupported Memory Media FRU. • All other encodings are reserved. Other Device Event Type values: • 00h = Device Event Sub-Type Not Reported. • All other encodings are reserved.
56h	2Ah	**Reserved**

8.2.10.2.1.4 Memory Sparing Event Record

The layout of a Memory Sparing Event Record is shown in Table 8-60.

Table 8-60. Memory Sparing Event Record (Sheet 1 of 2)

Byte Offset	Length in Bytes	Description
00h	30h	**Common Event Record**: See corresponding common event record fields defined in Section 8.2.10.2.1. The Event Record Identifier field shall be set to e71f3a40-2d29-4092-8a39-4d1c966c7c65 which identifies a Memory Sparing Event Record.
30h	1	**Maintenance Operation Class**: This field identifies the Class of a maintenance operation. See Table 8-125.
31h	1	**Maintenance Operation Subclass**: This field identifies the subclass of the maintenance operation together with the Maintenance Operation Class. See Table 8-125.
32h	1	**Flags** • Bit[0]: **Query Resources Flag**: If set, the CXL device is responding to a check for resources • Bit[1]: **Hard Sparing Flag**: If set, the host performed a sparing of permanent type; therefore, the sparing will be irreversible and will survive any Conventional Reset • Bit[2]: **Device Initiated**: If set, the CXL device was the originator of the sparing action; otherwise, the host initiated the sparing action • Bits[7:3]: **Reserved**
33h	1	**Result** • 00h = Success • 01h = Resources Exhausted • FFh = General Failure • All other encodings are reserved
34h	2	**Validity Flags**: Indicators of which fields are valid within the returned data. • Bit[0]: When set, the Channel field is valid • Bit[1]: When set, the Rank field is valid • Bit[2]: When set, the Nibble Mask field is valid • Bit[3]: When set, the Bank Group field is valid • Bit[4]: When set, the Bank field is valid • Bit[5]: When set, the Row field is valid • Bit[6]: When set, the Column field is valid • Bit[7]: When set, the Component Identifier field is valid • Bit[8]: When bit[8] and bit[9] are both set, the format of the Component Identifier field is governed by Table 8-56 • Bit[9]: When set, the Sub-channel field is valid • Bits[15:10]: **Reserved**
36h	6	**Reserved**
3Ch	2	**Spare Resource Available**: Specifies the remaining quantity of sparing resources available of the same granularity and at the same location in which the sparing occurred after the sparing operation is performed. This field is valid if the Result field is cleared to 00h. • 00h = No more resources are available • 01h = 1 more resource is available • 02h = 2 more resources are available • 03h = 3 or more resources are available • All other encodings are reserved
3Eh	1	**Channel**: Spared address as reported in the DRAM event for the errors (see Table 8-58).
3Fh	1	**Rank**: Rank associated with the spared address as reported in the DRAM event for the errors (see Table 8-58).

Table 8-60. Memory Sparing Event Record (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
40h	3	**Nibble Mask**: Mask of the final nibbles that are associated with the spared bits by the command (this could be greater than the requested spare nibble). Format for spared address as reported in the DRAM event for the errors. See Table 8-58.
43h	1	**Bank Group**: Bank group associated with the spared address as reported in the DRAM event for the errors (see Table 8-58).
44h	1	**Bank**: Bank associated with the spared address as reported in the DRAM event for the errors (see Table 8-58).
45h	3	**Row**: Row associated with the spared address as reported in the DRAM event for the errors (see Table 8-58).
48h	2	**Column**: Column associated with the spared address as reported in the DRAM event for the errors (see Table 8-58).
4Ah	10h	**Component Identifier**: Device-specific component identifier for the event. This may describe a field-replaceable sub-component of the device. This field is valid when bit[7] of the Validity Flags field is set. When bit[8] of the Validity Flags field is not set, the component identifier is device specific; otherwise, the component identifier format is defined by Table 8-56. This field should be provided for all subclasses of Memory Sparing and PPR.
5Ah	1	**Sub-Channel**: The sub-Channel field associated with the spared address (see Table 8-58). This field is valid when bit[9] of the Validity Flags field is set. This field should be provided if the Maintenance Operation Class field is Memory Sparing.
5Bh	25h	**Reserved**

8.2.10.2.1.5 Vendor Specific Event Record

The layout of a vendor specific event record is shown in Table 8-61.

Table 8-61. Vendor Specific Event Record

Byte Offset	Length in Bytes	Description
00h	30h	**Common Event Record**: See corresponding common event record fields defined in Section 8.2.10.2.1. The Event Record Identifier field shall be set to Vendor specific UUID representing the format of this vendor specific event.
30h	50h	**Vendor Specific Event Data**

8.2.10.2.1.6 Dynamic Capacity Event Record

All Dynamic Capacity event records shall set the Event Record Severity field in the Common Event Record Format to Informational Event. All Dynamic Capacity related events shall be logged in the Dynamic Capacity Event Log. This definition of the event is for an LD-FAM DCD. See Table 8-62.

Table 8-62. Dynamic Capacity Event Record (Sheet 1 of 2)

Byte Offset	Length in Bytes	Description
00h	30h	**Common Event Record**: See corresponding common event record fields defined in Section 8.2.10.2.1. The Event Record Identifier field shall be set to ca95afa7-f183-4018-8c2f-95268e101a2a which identifies a Dynamic Capacity Event Record.
30h	1	**Dynamic Capacity Event Type**: Identifies the type of Dynamic Capacity event that occurred. • 00h = Add Capacity. The device is requesting add of Dynamic Capacity for the host and the Dynamic Capacity Extent field in this structure specifies the added capacity. The host shall respond with a single Add Dynamic Capacity Response command. This event shall only be reported to the host. • 01h = Release Capacity. The device is requesting release of Dynamic Capacity from the host and the Dynamic Capacity Extent field in this structure specifies the released capacity. The host may respond with zero or more Release Dynamic Capacity requests. This event shall only be reported to the host. • 02h = Forced Capacity Release. The device has forcefully released Dynamic Capacity from the host and the Dynamic Capacity Extent field in this structure specifies the released capacity. The host may respond with zero or more Release Dynamic Capacity requests. This event shall only be reported to the host. • 03h = Region Configuration Updated. The configuration of a DC Region has been updated, as indicated by the Updated Region Index field. This event shall only be reported to the host. • 04h = Add Capacity Response. The host has responded to the Add Capacity event and the Dynamic Capacity Extent field in this structure specifies the capacity accepted by the host. This event shall only be reported to the FM. • 05h = Capacity Released. The host has released capacity and the Dynamic Capacity Extent field in this structure specifies the released capacity. This event shall only be reported to the FM. • All other encodings are reserved.
31h	1	**Validity Flags** • Bit[0]: When set, the Number of Available Extents field and the Number of Available Tags field are valid • Bits[7:1]: **Reserved**
32h	2	**Host ID**: For an LD-FAM device, the LD-ID of the host interface that triggered the generation of the event. For a GFD, the PBR ID (PID) of the Edge USP of the host that triggered the generation of the event. This field is valid only for events of type Add Capacity Response and Capacity Released.
34h	1	**Updated Region Index**: Index of region whose configuration was updated. This field is only valid for events of type Region Configuration Updated.
35h	1	**Flags** • Bit[0]: **More**: — 0 = This is the last (or the only) record associated with this event. — 1 = The next Event Record is also associated with this event, providing an additional extent. Records grouped this way shall be continuous in the Event Log, with no unrelated records between them, and shall contain the same Dynamic Capacity Event Type. • Bits[7:1]: **Reserved**

Table 8-62. Dynamic Capacity Event Record (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
36h	2	**Reserved**
38h	28h	**Dynamic Capacity Extent**: See Table 8-63.
60h	18h	**Reserved**
78h	4	**Number of Available Extents**: The remaining number of extents that the device supports, as defined in Section 9.13.3.3.
7Ch	4	**Number of Available Tags**: The remaining number of Tag values that the device supports, as defined in Section 9.13.3.3.

For simplicity, Dynamic Capacity Event Records with Dynamic Capacity Event Type field set to Add Capacity is referred to as Add Capacity Event Record. The term Release Capacity Event Record is used elsewhere in the document to refer to Dynamic Capacity Event Records with Dynamic Capacity Event Type set to Release Capacity. The term Forced Release Capacity Event Record is used to refer to Dynamic Capacity Event Records with Dynamic Capacity Event Type set to Forced Capacity Release.

Table 8-63. Dynamic Capacity Extent

Byte Offset	Length in Bytes	Description
00h	08h	**Starting DPA**: The start address of the first block of extent data. Always aligned to the Configured Region Block Size returned in Get Dynamic Capacity Configuration. The extent described by the Starting DPA and Length shall not cross a DC Region boundary. • Bits[5:0]: **Reserved** • Bits[63:6]: **DPA[63:6]**
08h	08h	**Length**: The number of contiguous bytes of extent data. Always a multiple of the configured Region Block Size returned in Get Dynamic Capacity Configuration. The Length shall be > 0. The extent described by the Starting DPA and Length shall not cross a DC Region boundary.
10h	10h	**Tag**: Optional opaque context field utilized by implementations making use of the Dynamic Capacity feature. Within sharable regions, all extents targeting the same range this field is required and all Tag values shall be identical for all hosts sharing the same extent. See Section 7.6.7.6.5 for how the Tag field is utilized while setting up sharing.
20h	02h	**Shared Extent Sequence**: The relative order that hosts place this extent within the virtual address space. Each instance of shared memory is presented to the sharing hosts with the same Tag value. The Shared Extent Sequence is used so that the same offsets reach the same data accessed by all hosts. For extents that describe non-shareable regions, this field shall be 0. For extents describing shareable regions this field shall be within the range of 0 to n-1 where n is the number of extents, with each value appearing only once. Extents may appear in any order and may contain gaps between them in DPA space.
22h	06h	**Reserved**

8.2.10.2.2 Get Event Records (Opcode 0100h)

Retrieve the next event records that may exist in the device's requested event log. This command shall retrieve as many event records from the event log that fit into the mailbox output payload. The device shall set the More Event Records indicator if there are more events to get beyond what fits in the output payload. Devices shall return event records to the host in the temporal order the device detected the events in. The event occurring the earliest in time, in the specific event log, shall be returned first.